Financial Data Mining and Analysis

Chapter 3: Time value of money

Agenda: Chapter 3: Time value of money

- Review for several formula
  - Future value function
  - Present value function
  - Present value of perpetuity
  - Present value of a growing perpetuity
  - Present value of annuity
  - Future value of annuity
  - Excel sign conversion
  - Excel corresponding functions
- Introduction to the numpy_financial module
- Decision Rules
  - NPV and its rule
  - IRR and its rule
  - Payback period and its rule

Timeline

Assume that we borrow $100 for two years
the annual interest rate is 10%.

Below is a timeline with cash flows.

```
100               10               10+100
   |-------------|-------------|
   0             1             2
```

We receive $100 at time zero and pay $10
for the next two years, plus $100 at the end of
year two.

Future value function

Calculations?

$$FV = 100 \times (1 + 0.0123)^5$$
$$FV \approx 106.30$$

The present value for a given future value is

fv= pv*(1+R)^n

where FV is the future value

R is the period rate

n is the number of periods

Note: R and n should be consistent, i.e., with the same frequencies.

For example, what is the future value if we deposit $100 today with an annual rate of 1.23% for 5 years?

Present value function

The present value for a given future value is

$$pv = \frac{fv}{(1+R)^n}$$

where FV is the future value

R is the period rate

n is the number of periods

Note: R and n should be consistent, i.e., with the same freq.

For example, what is the present value of $100 received at the end of year 5 with an annual discount of 5%?

Present value of perpetuity

Definition of a perpetuity: same cash flows at the same intervals forever

The present value for a given future value is

$$\boldsymbol{pv(perpetuity) = \frac{C}{R}}$$

where C is the cash flow

R is the period rate

Note: the first cash flow happens at the end of the first period.

For example, what is the present value of $100 received every year forever with a discount rate of 10% ? Assume that the first $100 received at the end of the first year.

Present value of annuity

Definition of annuity: same cash flows at the same intervals for n periods

The present value for a given future value is

$$pv(annuity) = \frac{c}{R}\left(1 - \frac{1}{(1+R)^n}\right)$$

where C is the cash flow

R is the period rate

n is the number of periods

Note: R and n should be consistent, i.e., with the same frequencies.

The first cash flows happens at the end of the first period.

For example, what is the present value of $100 received at the end of year for the next 10 years with an annual discount of 5%?

Future value of annuity

Definition of annuity: same cash flows at the same intervals for n periods

The present value for a given future value is

$$\boldsymbol{fv(annuity) = \frac{C}{R}[(1+R)^n - 1]}$$

where C is the cash flow

R is the period rate

n is the number of periods

Note: R and n should be consistent, i.e., with the same frequencies.

The first cash flow happens at the end of the first period.

For example, what is the future value of $100 received at the end of the year for the next 10 years with an annual rate of 5%?

perpetuity due and annuity due

For normal perpetuity or annuity, each cash flow would happen at the end of each period.

For perpetuity due or annuity due, each cash flow would happen at the beginning of each period.

There are two ways to calculate present (future) value of annuity due.

Cash flows of a growing annuity

```
 c          c(1+g)     c(1+g)^2    c(1+g)^(n-1)    0
 |-----------|-----------|------   ---|------------|
 0           1           2            n-1          n
```

```
The following pattern is for our formula
```

```
             c          c(1+g)     c(1+g)^2     c(1+g)^(n-1)
 |-----------|-----------|------   ---|------------|
 0           1           2            n-1          n
```

In-class exercise #1: Excel pv() function

- What is the present value of a given future value
- Write a Python function to mimic the Excel function

Excel sign convention

There are two ways to explain the Excel sign convention.

1) Accounting approach: using cash inflow and cash output.

Example #1: we deposit $100 for 1 year. If the interest rate is 10%, how much is the value 1 year later?

When deposit $100, it is a cash output

=fv(0.1,1,0,-100) => 110 (a positive value)

Excel sign convention (2)

2) Excel is designed this way

positive future value -- > negative present value

negative future value -- > positive present value

positive present value -- > negative future value

negative present value -- > positive future value

A general formula

Present value is the sum of the present value of a future value plus the present value of annuity

$$pv = pv(fv) + pv(annuity)$$

$$pv = \frac{fv}{(1+R)^n} + \frac{c}{R}\left(1 - \frac{1}{(1+R)^n}\right)$$

Given 4 values, we can calculate number 5

In the following formula, we have five values: pv, fv, R, c, and n.

$$pv = \frac{fv}{(1+R)^n} + \frac{c}{R}\left(1 - \frac{1}{(1+R)^n}\right)$$

For a set of 4 values, we can calculate the fifth unknow.
For Excel, we have the pv(), fv(), pmt(), nper() and rate() functions.

For Python, we use the numpy_financial

$$pv = \frac{fv}{(1+R)^n} + \frac{c}{R}\left(1 - \frac{1}{(1+R)^n}\right)$$

For a set of 4 values, we can calculate the fifth unknow.

In-class exercise #2: install numpy_financial

- Task #1: install numpy_financial
- Task #2: how many functions inside the package
- Task #3: find the usage of one function such as rate
- Task #4: test this function
- Task #5: compare it with the corresponding Excel function

numpy_financial Python module

```
import numpy_financial as npf
a=dir(npf)
print(a)
```

```
['__builtins__', '__cached__', '__doc__',
'__file__', '__loader__', '__name__',
'__package__', '__path__', '__spec__',
'__version__', '_financial', 'fv', 'ipmt',
'irr', 'mirr', 'nper', 'npv', 'pmt', 'ppmt',
'pv', 'rate']
```

The numpy_financial.pv() function

Note: numpy_financial.pv() functions mimics Excel pv() function

```
import numpy_financial as npf
help(npf.pv)
Help on function pv in module
numpy_financial._financial:
pv(rate, nper, pmt, fv=0, when='end')
    Compute the present value.

npf.pv(0.1,1,0,110)
Out[52]: -99.99999999999999
```

In-class exercise #3: compare pv() functions for Excel and numpy_financial

- For Excel : =pv(

```
# For Python
import numpy_financial as npf
```

Present value of a growing annuity

Definition of annuity: same cash flows at the same intervals for n periods

The present value for a given future value is

$$pv(growing\ annuity) = \frac{c}{R}(1 - \frac{(1+g)^n}{(1+R)^n})$$

where C is the cash flow

R is the period rate

g is <u>the growth rate</u>

n is the number of periods

Note: R and n should be consistent, i.e., with the same frequencies.

The first cash flows happens at the end of the first period.

For example, what is the present value of a growing 10-year annuity with the first cash flow of $100 received at the end of the first year? The cash flows will grow at a constant 2% per year with an annual discount of 15%.

In-class exercise #4: Python program for the present value of a growing annuity

- Function name could be pv_growing_annuity
- Add comments or help
- Add several examples
- Add at least two ways to input parameters

Future value of a growing annuity

Definition of annuity: same cash flows at the same intervals for n periods

The present value for a given future value is

$$fv(growing\ annuity) = \frac{C}{R}[(1+R)^n - (1+g)^n]$$

where C is the cash flow

R is the period rate

g is the growth rate

n is the number of periods

Note: R and n should be consistent, i.e., with the same frequencies.

The first cash flow happens at the end of the first period.

For example, what is the future value of a growing 10-year annuity with the first cash flow of $100 received at the end of the first year? The cash flows will grow at a constant 2% per year with an annual discount of 15%.

In-class exercise #5: NPV using both Excel npv() and numpy_financial.npv() functions

- Test a few sets of values

In-class exercise: pv(perpetuity with the first cash flow happens at the end of kth period

The formula is given below.

$$pv(perpetuity, 1st\ CF\ at\ k) = \frac{1}{(1+R)^{k-1}} * \frac{C}{R}$$

NPV's definition

NPV stands for Net Present Value

NPV= pv(all cash inflows) – pv(all cash outflows)

If defining cash inflows as positive and cash outflows as negative, then we have the following simple formula.

NPV= pv(all cash flows)

NPV Rule

- For given project, we calculate its NPV first, then apply the following rule.

If NPV >0, we accept the project

<0, we reject the project

Calculate IRR of the following cash flows

- cashflows=[-100,50,60,70]
- R (cost of capital) =0.12

One NPV example

Here is an example. The initial investment is $100. The cash inflows in the next five years are $50, $60, $70, $100, and $20, starting from year one. If the discount rate is 11.2%, what is the project's NPV value? Since only six cash flows are involved, we

1) do the calculation manually
2) using Excel npv() function
3) use the **numpy_financial.npv()** function

Calculate the NPV of the following cash flows

- cashflows=[-100,50,60,70]

IRR's definition

- IRR stands for Internal Rate of Return.
- The discount rate makes its corresponding NPV zero.

- **NPV(when R is IRR) = 0**

IRR Rule

- For a project, we calculate its IRR first, then we apply the following rule.

If IRR > R, we accept the project

< R, we reject the project

where R is the cost of capital of the firm.

Payback period

- The definition of Payback is the number of years need to recover the initial investment.

- Two ways:
  - The first method does not consider the Time value of Money
  - The second method considers the Time value of money

Decision rule based on the payback period

- We calculate the payback of our project first. Then, we apply the following rule.

If $T < T_c$, we accept the project

$T > T_c$, we reject the project

where T is the payback of our project and Tc is the maximum payback period your company allows.

Without considering the time value of money

- Assume that our initial investment is 100
- Our next five years cash flows are 50, 30, 30, 20, and 25.
- Our payback period is 2 + 20/30 = 2.67 years

Considering the time value of money

- Assume that our initial investment is 100
- Our next five years cash flows are 50, 30, 30, 20, and 25.
- In addition, the required rate of return is 12% per year.
- What is the payback period of this project?

graphical presentation of NPV profile with two IRRs

```
import numpy as np
import matplotlib.pyplot as plt
Cashflows=[504,-432,-432,-432,832]
```

[see the textbook on page 131]